Methodology

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Cross-Modal Bayesian Reward Filter

- Reward system formula
  - $$r_{total} = r_{env} + \left(\frac{\alpha}{\alpha + \beta}\right) \cdot r_{human}$$

    If $sign(r_{human}) = sign(\Delta\Phi)$, then $\alpha \leftarrow \alpha + 1$

    If $sign(r_{human}) \neq sign(\Delta\Phi)$, then $\beta \leftarrow \beta + 1$
- When the formula is applied in the system (diagram + written explanation)
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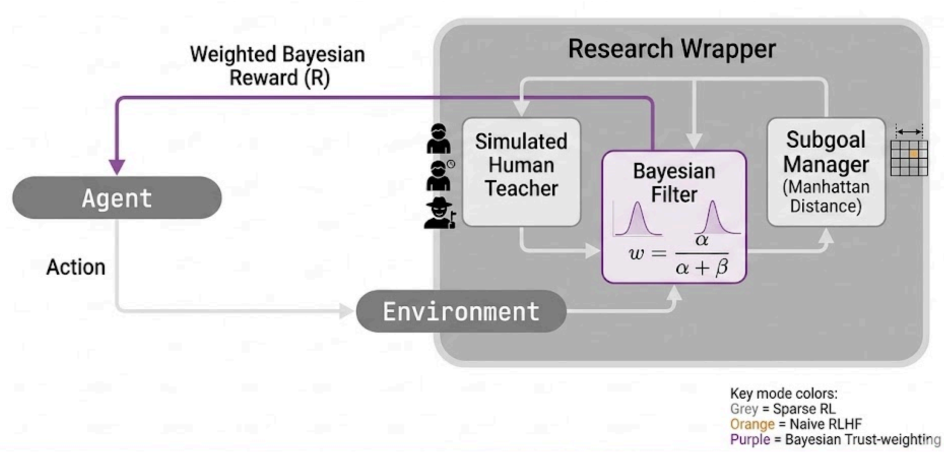
  - 

- Compared with Sparse and Naive baselines

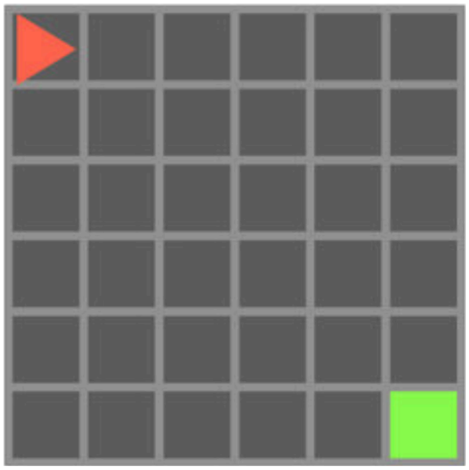
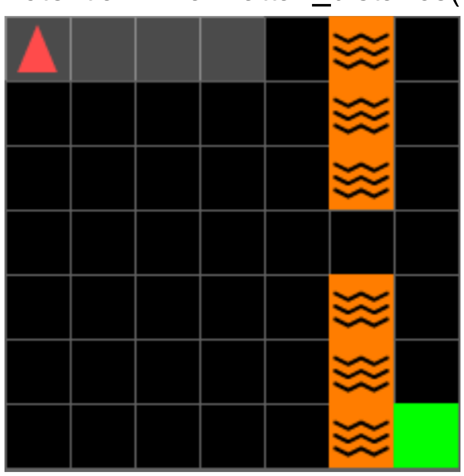
Model Hyperparameters

- Base RL hyperparameters use Stable-Baselines3 defaults, which are standard for MiniGrid-scale tasks. All conditions share identical hyperparameters, preserving internal validity of comparisons.
  - Total time-steps (# times agent takes action per run): 15,000 steps/run
  - Epochs (# times agent optimizes over a batch of data): 10
  - Learning rate: 3 * 10^ (-4)
  - Optimizer: Adam
  - Evaluation episodes (sample size to calculate success rate): 20
- Bayesian Reward Filter hyperparameters:
  - Bayesian priors (initial values):
    - $$\alpha_{init} = 1.0, \beta_{init} = 1.0$$
  - human_magnitude = 0.1 (scale of human feedback signal +- 0.1)

Testing Reward Filter

- Table showing: environments x humans x agents

Environments

- MiniGrid-Empty-8x8-v0
  - Goal: reach green tile in an empty room
  - Potential: −manhattan_distance(agent, goal)
  - 
  - https://www.researchgate.net/publication/357231210/figure/fig2/AS:11431281207831088@1701351736280/Empty-environment-from-MiniGrid-We-consider-three-environments-from-MiniGrid-in-this.jpg
- MiniGrid-LavaGapS7-v0
  - Goal: reach goal without stepping on lava strip
  - Potential: −manhattan_distance(agent, goal) − lava_adjacency_penalty
  - 
  - https://minigrid.farama.org/environments/minigrid/CrossingEnv/
- MiniGrid-DoorKey-8x8-v0
  - Goal: pick up key, unlock door, reach goal
  - Potential: staged - distance to key (then door, then goal)

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- https://media.springernature.com/lw685/springer-static/image/art%3A10.1007%2Fs00521-025-11340-0/MediaObjects/521_2025_11340_Fig8_HTML.png

Types of Humans

- Many humans used to stimulate different types of noisy input the model may receive
- StochasticHuman: fixed-rate noise baseline
  - StochasticLow (0.1 noise level) and StochasticHigh (0.8 noise level)
- AdversarialHuman: always lies (worst-case attacker)
- DistanceNoiseHuman: confusion scales with distance from goal
- FatigueHuman: Noise grows linearly from 0% to 80% over 50,000 steps
- RealisticHuman: combines distance, fatigue, and base noise

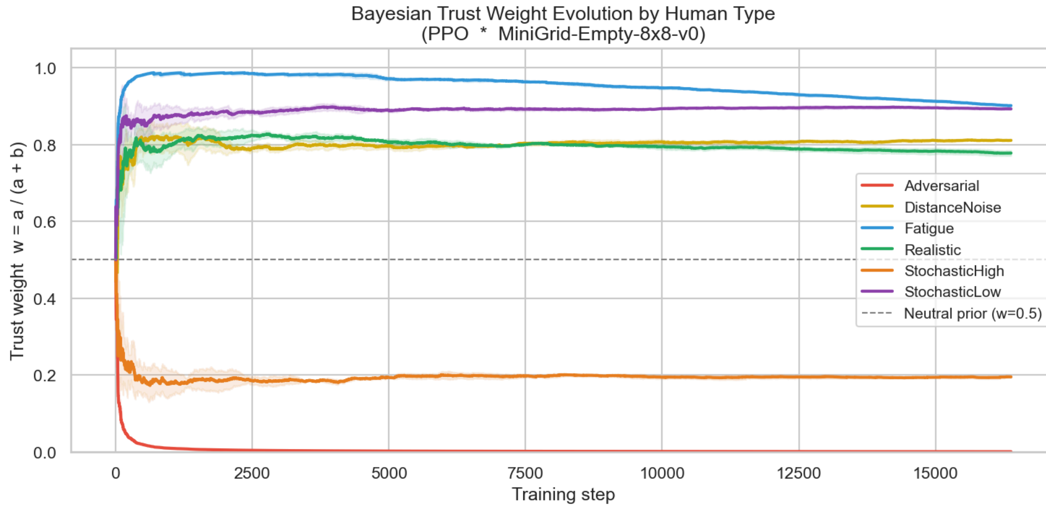
Agents

- PPO and DQN
- Both chosen to represent on-policy and off-policy learning

Results

Results

- Trust Weight Changes Over Training by Human



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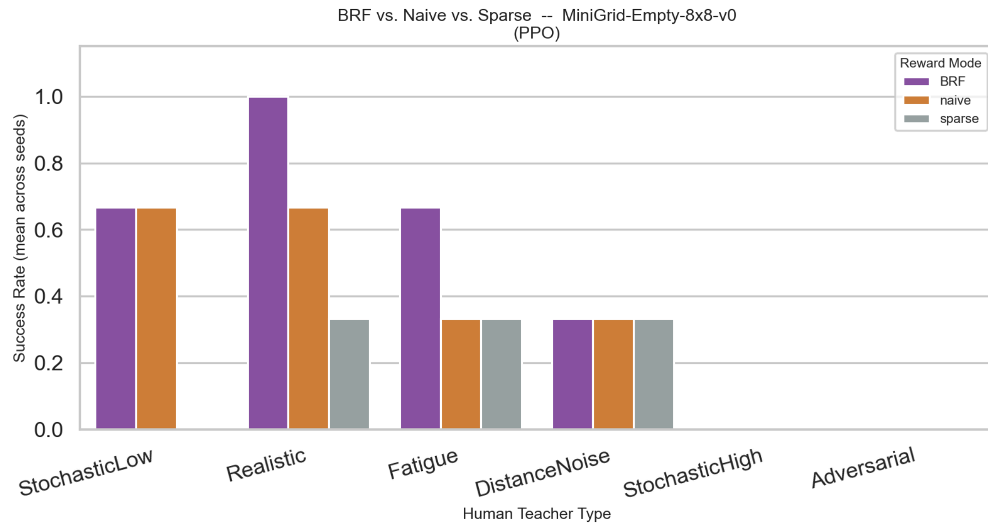
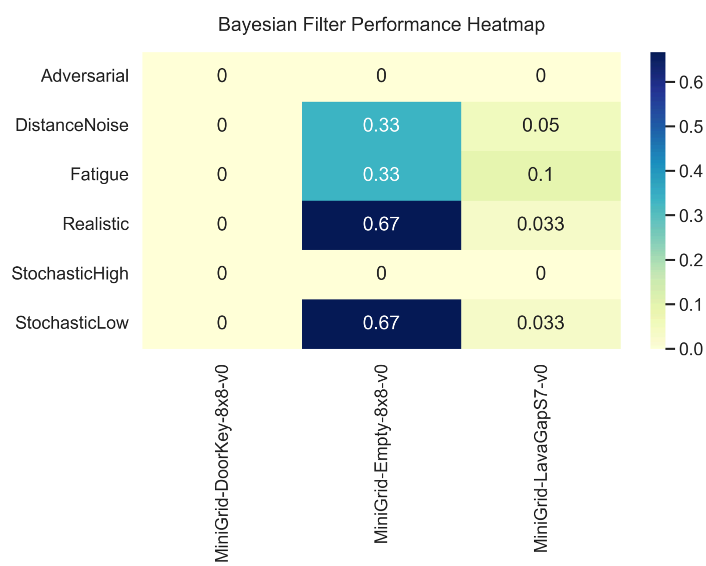
  - Weight decreasing as human gets worse for Fatigue scenario, but trusts it highly at beginning.
  - Does not trust the adversarial human at all (w=0)
  - StochasticHigh has low weight, but still some (around 0.2)
  - Although trust stabilizes in most scenarios, the Fatigue and Realistic curves show a slight decline at the showing that the score adapts to the human as time passes
  - Distance noise only shows a slight increase in score as the agent gets closer. Increase is not monumental because the grids are pretty small to begin with
- PPO table and graph showing success across all cases on Empty MiniGrid

Human	**Sparse**	**Naive**	**Bayesian**
Adversarial	0.000	0.000	0.000
DIstanceNoise	0.333	0.333	0.333
Fatigue	0.333	0.333	0.666
Realistic	0.333	0.667	1.0
Stochastic High	0.000	0.000	0.000
Stochastic Low	0.000	0.667	0.667

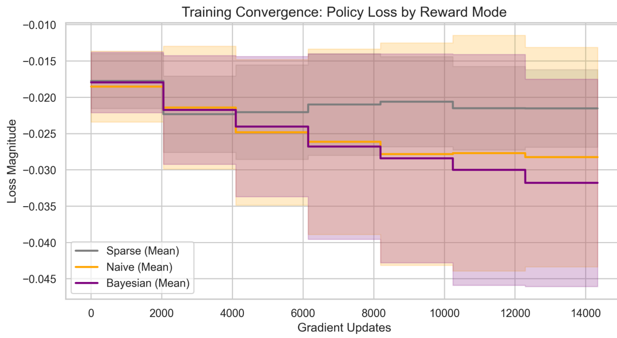
- 

- BRF achieves perfect success (1.0) under Realistic noise, vs. 0.667 Naive and 0.333 Sparse
- BRF better in Fatigue/Realistic
- All methods tie in DIstance Noise and Stochastic Low
- Adversarial and StochasticHigh fail universally (human reward too corrupt to get any useful information, but environment reward too sparse)

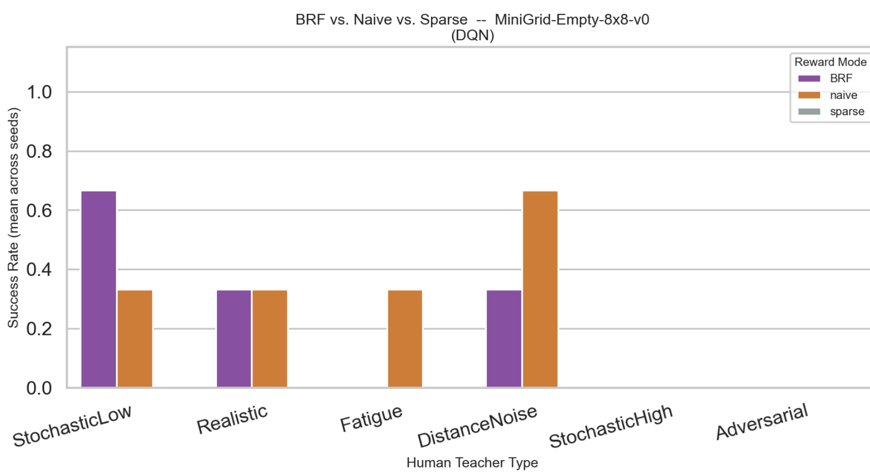
- PPO Heatmap



  - Empty-8x8 is the only environment where any method succeeds

- 

    - Policy loss more negative in PPO, so the policy update is larger. Bayesian Reward filter becomes negative fastest (reaching -0.031) while Sparse barely moves, showing the reward signal is more useful and driving better policy updates
    - Large confidence interval for BRF, because it accounts for all humans (ncluding less successful humans like adversarial)
- DQN proved to be too complex and the human/environment reward was too sparse. It is off-policy, meaning it relays past experience. This means corrupted human feedback from earlier in training continues to be used, even though w currently downplays that human. In the future, look at cases where this Bayesian Reward Filter can be used with other agents when there is more reward



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Environment	Human	Sparse	Naive	Bayesian
Doorkey	All types	0.000	0.000	0.000
Empty	Adversarial	0.000	0.000	0.000
	DistanceNoise	0.000	0.667	0.333
	Fatigue	0.000	0.333	0.000

	Realistic	0.000	0.333	0.333
	StochasticHigh	0.000	0.000	0.000
	StochasticLow	0.000	0.333	0.667
LavaGap	Fatigue	0.000	0.083	0.200
	Realistic	0.000	0.183	0.067
	(others)	0.000	~0	~0-0.1

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- Naive (0.667) beats Bayesian (0.333) for DQN.
- Bayesian beats Naive for Stochastic Low
- DQN's sparse baseline is 0.0 everywhere. DQN can't learn from sparse rewards alone in these environments at 15k steps

Analysis

Analysis

- Bayesian filter matched or exceeded sparse and naive baseline results on the empty minigrid with PPO
  - Increased by 0.333 for fatigue and realistic human noise levels
  - Meaning: Low-compute way of improving training and therefore final results
- Trust score evolution shows model adapts based on the human noise compared to the environment
- Where it did not succeed
  - LavaGap saw minor success (0.05–0.10). Filter provides some signal but is too little to help solve environment
  - DoorKey is always 0 success out of the 20 trials. Reward structure may be too sparse to solve in 15,000 steps. Not necessarily a failure of the filter itself
  - BRF saw worse performance in some cases compared to sparse/naive when using DQN agents. Could be because DQN is off-policy and continues to use past human feedback (even if the trust score has learned that the human is not trustworthy). How can the BRF approach be adapted to a DQN agent?
- Stochastic Low succeeded at different levels for both PPO and DQN, but Stochastic High failed (0.000 on evaluation runs) for both. Shows there could be a threshold between recoverable versus unrecoverable noise
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